Finite-dispersion temperature corrections in stationary quartic-superluminal Hawking scattering

Tatsuhiko Takimoto

Independent researcher

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Abstract

We examine the frequency dependence of a temperature defined by the ratio of Bogoliubov coefficients for a stationary, one-dimensional flow with quartic superluminal dispersion. Two independently developed numerical methods, stabilized spatial integration and a global Chebyshev boundary-value method, give compatible scattering data within the recorded comparison precision. For 19 smooth profiles, the stored extrapolations of the ω^2/Λ^2 coefficient agree with a local derivative expression to an absolute difference below 1.3×10^{-3}. This expression is a short consequence of the Leonhardt–Robertson phase-integral formula; the numerical comparison concerns its applicability to the full four-mode wave equation. The results differ from the coefficient obtained by directly applying Eq. (133) of Del Porro, Liberati and Schneider. The domains and orders of approximation must be reconciled before interpreting that difference as a disagreement of equivalent predictions. For the low-frequency temperature offset we apply the zero-frequency matching procedure of Coutant and Weinfurtner to this wave equation. Under stated regularity, regularization and matching assumptions, the $\omega \to 0$ limit of the ratio temperature reduces, without approximation in the dispersion scale Λ, to a Wronskian and two integrals of one solution of a second-order equation. A formal expansion about the horizon turning point then gives the relative offset $\kappa\, g'''(0)/(12\Lambda^2)$, where g expresses v' as a function of $1 + v$. This is a formal asymptotic calculation: global matching and a remainder bound are not proved. It is supported numerically: the zero-frequency reduction agrees with extrapolated $\omega \to 0$ scattering limits within 1.3×10^{-10} in 28 cases, and predictions recorded before any zero-frequency or scattering calculation for two new profile families agree with the offsets extrapolated from those limits within 1.1×10^{-7}. A separately written scattering code reproduces the predicted offset for four profiles, two of them new, within 1.1×10^{-7}.

1 Question and relation to previous work

The leading Hawking temperature is robust in many dispersive models; for example, Corley and Jacobson analysed horizon mode conversion with high-frequency dispersion [1], and Unruh and Schützhold examined the conditions under which the effect is universal [2]. Finite-dispersion deviations, in turn, contain information about the background flow and the observable used to define a temperature. We ask which coefficient is measured by full stationary scattering when the low-frequency offset and the frequency curvature are separated. With sound speed set to one, the horizon is at $v(0) = -1$, its slope is $\kappa = v'(0) > 0$, and $T_{\mathrm{H}} = \kappa/(2\pi)$; $\hbar = k_B = 1$ throughout.

The analysis builds on distinct contributions. Macher and Parentani developed the three-channel scattering description and separated an effective low-frequency temperature from spectral curvature [3]. Coutant, Parentani and Finazzi developed analytical connection and momentum-space methods [4]. Leonhardt and Robertson expressed the temperature through an inverse-flow phase integral [5]. Moreno-Ruiz and Bermudez applied such methods to nonlinear profiles and compared analytical and numerical temperatures [7]. Himemoto and Tanaka studied finite-dispersion corrections involving higher background derivatives [6]. Del Porro, Liberati and Schneider developed a tunneling method for analogue Hawking quanta, for flows with and without horizons and with sub- or superluminal dispersion [8], and then connected the tunneling and Bogoliubov descriptions through an effective horizon [9]. Eq. (133) of the latter work supplies the comparison coefficient in this report. Coutant and Weinfurtner showed how to read low-frequency scattering coefficients from a zero-frequency solution of a second-order equation and its first-order frequency correction [10, 11]; our low-frequency calculation follows that procedure.

The numerical quantities of interest are the coefficients in

$$\frac{T(\omega,\Lambda)}{T_{\mathrm{H}}} - 1 \simeq \frac{b_0\kappa^2 + c\omega^2}{\Lambda^2} + \cdots . \tag{1}$$

Here $\simeq$ specifies an asymptotic interpretation of finite-range fits, rather than a uniform error theorem. We distinguish the formal phase-integral value of c, the measured full-scattering coefficients, and the low-frequency coefficient b_0, which Section 7 obtains by a formal asymptotic calculation. The calculation is a comparison of these quantities, rather than a claim that dispersive temperature corrections, the phase-integral method or the zero-frequency method are new.

2 Wave equation, channels, and numerical methods

For a prescribed stationary velocity $v(x)$, the model is

$$(\partial_t + \partial_x \circ v)(\partial_t + v\partial_x)\phi = \partial_x^2\phi - \Lambda^{-2}\partial_x^4\phi. \tag{2}$$

The first factor acts as $\partial_t + \partial_x(v\,\cdot)$. For $\phi = e^{-i\omega t}f(x)$, Eq. (2) becomes

$$\Lambda^{-2}f'''' = (\omega^2 + i\omega v')f + (2i\omega v - 2vv')f' + (1-v^2)f''. \tag{3}$$

At constant velocity the wave numbers obey

$$(\omega - vk)^2 = k^2 + \frac{k^4}{\Lambda^2}. \tag{4}$$

Thus ω is the conserved laboratory frequency; the comoving frequency is $\Omega_c = \omega - vk$. This distinction matters when comparing notations between analytical treatments.

The left asymptotic region is supersonic and has four real roots below the channel cutoff. On the subsonic right, two roots propagate and two are evanescent. The exponentially increasing right-hand solution is excluded. The incoming channels are the right-hand co-propagating channel, and the positive- and negative-norm high-wave-number channels on the left. The outgoing channels are the right-hand counter-propagating mode u and two left-hand modes. Norm signs follow $\operatorname{sgn}\Omega_c$; directions follow the laboratory group velocity

$$v_g = \frac{v\Omega_c + k + 2k^3/\Lambda^2}{\Omega_c}. \tag{5}$$

Writing $D = -i\omega + v\partial_x$, the conserved bilinear current is

$$\begin{aligned} J(f_1,f_2) = {} & v[f_1^* Df_2 - f_2(Df_1)^*] - (f_1^* f_2' - f_2 f_1^{*\prime}) \\ & + \Lambda^{-2}[f_1^* f_2''' - f_1^{*\prime} f_2'' + f_1^{*\prime\prime} f_2' - f_1^{*\prime\prime\prime} f_2]. \end{aligned} \tag{6}$$

For $f = ae^{ikx}$, $(i/2)J(f,f) = \Omega_c v_g |a|^2$. Normalizing propagating amplitudes with $\sqrt{|\Omega_c v_g|}$ gives a 3×3 scattering matrix with $S^\dagger \eta S = \eta$, $\eta = \mathrm{diag}(1,1,-1)$, after ordering by norm sign. The outgoing u row defines

$$|\alpha|^2 = |S_{u,+}|^2, \qquad |\beta|^2 = |S_{u,-}|^2, \qquad G = |S_{u,v}|^2, \qquad |\alpha|^2 - |\beta|^2 = 1 - G, \tag{7}$$

and our principal observable is

$$T_{\mathrm{ratio}} = \frac{\omega}{\log(|\alpha|^2/|\beta|^2)}. \tag{8}$$

One numerical method integrates a physical solution basis from right to left with an adaptive eighth-order integrator, repeatedly removing the evanescent component from the propagating basis vectors. The other solves Eq. (3) on multiple Chebyshev elements as one sparse linear boundary-value problem. Continuity of all four components is imposed at element boundaries, and the three separate incoming-mode problems share a sparse factorization. The standard elements have degree 28; selected checks use degree 36, shorter elements and a longer domain. Their independent development provides a check on spatial integration versus global collocation for the symmetric-tanh comparison set; the additional profiles below use the boundary-value method.

3 What is computed, and how

All numerical values use $\kappa = 1$; frequencies and cutoffs are therefore in units of κ. The data cover the following finite calculations.

Quantity	Method and range	Status of the comparison
Scattering cross-check	Symmetric tanh; $D = 0.2, 0.5, 0.8, 1$; $\Lambda = 5, 10, 20, 40$; sampled $0.05 \le \omega \le 1.2$	93 numeric table entries, 64 distinct points; 3 cutoff entries
Smooth-profile scans	19 profiles; $\Lambda = 10, 20, 40, 80$; 9 frequencies from 0.025 to 0.8	746 successful solves including extensions; 2 above-cutoff cases excluded
Large-cutoff extension	M_3, M_5, B_1, B_2; $\Lambda = 160, 320$; 8 frequencies through 0.6	Refined-mesh checks through $\Lambda = 160$; none at 320
Coefficient extraction	Quartic fit in ω through 0.4; sextic-fit comparison through 0.6	Three-largest-cutoff extrapolation in Λ^{-2} and Λ^{-4}
Zero-frequency reduction (Sec. 7.2)	Second-order equation (21) integrated with an adaptive eighth-order method, relative tolerance 10^{-12}; 7 profiles, $\Lambda = 10$–320 (10–160 for two)	Exact in Λ under A1, A2; checked against scattering limits in 28 cases
Scattering limits $\omega \to 0$	Boundary-value method at $\omega = 0.02, 0.01, 0.005, 0.0025$; T_{ratio} cubic in ω^2, $\omega\lvert\beta\rvert^2$ and G cubic in ω; $\Lambda = 10$–80	Four-point extrapolation; b_0 from $\Lambda = 20, 40, 80$ in Λ^{-2}, Λ^{-4}
Local expansion (Sec. 7.3)	Polynomial-times-Airy recursion in exact rational arithmetic through ε^6, i.e. Λ^{-4}	Formal; termwise integration under A3; remainder not bounded
Prediction-first test (Sec. 7.5)	Two new families; b_0, b_1 recorded from the local formulas before any solve	Three criteria fixed in advance; all met
Independent numerical check (end of this section)	Separately written scattering code; $T_{0.5}, B_4, N_1, N_2$; $\Lambda = 10$–160, $\omega = 0.0025$–0.02	Predicted b_0 reproduced within 1.1×10^{-7}; numerical agreement, not a proof of A1–A3

The nine scan frequencies are $0.025, 0.05, 0.1, 0.15, 0.2, 0.3, 0.4, 0.6, 0.8$. At each cutoff the stored primary fit of $\Delta = T_{\mathrm{ratio}}/T_{\mathrm{H}} - 1$ is

$$\Delta = a_0(\Lambda) + a_2(\Lambda)\omega^2 + a_4(\Lambda)\omega^4, \quad c(\Lambda) = \Lambda^2 a_2(\Lambda), \quad b_0(\Lambda) = \Lambda^2 a_0(\Lambda). \tag{9}$$

The extrapolated values $c_{\mathrm{ext}}, b_{0,\mathrm{ext}}$ are the intercepts of the stored fits in $1/\Lambda^2$ and $1/\Lambda^4$ using the three largest available cutoffs. These three-point extrapolations supply estimates, not confidence intervals. No new scattering solve or refit is needed to produce the tables and figures in this report. Provenance, transcription checks and the input-to-profile mapping are in the separate verification record.

Status of the low-frequency derivation. The formula for b_0 in Section 7 is the result of a formal asymptotic calculation, supported by the numerical and prediction-first checks listed above; it is not a theorem. It rests on three assumptions, stated precisely in Section 7.1: regularity of the low-frequency expansion together with its matching to the asymptotic waves (A1); the treatment of oscillatory tails by Abel continuation, which fixes two integration constants (A2); and the replacement of global integrals and of the travelling-wave normalization by their local turning-point expansions (A3). The main gaps are therefore the global connection of the local solution to the scattering data, including the normalization of the incoming travelling component, and a bound on the remainder. The quoted $O(\Lambda^{-4})$ denotes the next order of the formal expansion, not an error estimate.

Profiles. The symmetric families are $T_D : v = -1 + D\tanh(x/D)$, with $D = 0.5, 1$, and $F_m : v = -1 + \int_0^x e^{-z^{2m}}\,\mathrm{d}z$, with $m = 1, 2, 3, 4$. Define a normalized shifted step by

$$H_{q,t}(x) = \frac{\tanh(qx - \operatorname{arctanh} t) + t}{q(1 - t^2)}. \tag{10}$$

Then $S_{q,t}$ has $v = -1 + H_{q,t}$, with $(q, t) = (2, \pm 1/8), (2, \pm 1/4), (4, \pm 1/8)$. The remaining profiles have $v = -1 + \sum_i h_i H_{q_i,t_i}$ and $\sum_i h_i = 1$:

Profile	Triples (h_i, q_i, t_i)
A_+	$(0.5, 2, 0.25)$; $(0.5, 1, -0.5)$
A_-	$(0.5, 2, -0.25)$; $(0.5, 1, 0.5)$
M_3	$(0.3, 3, 0.2)$; $(0.7, 1, -0.1)$
M_4	$(0.6, 1.5, -0.3)$; $(0.4, 3, 0.1)$
M_5	$(0.5, 3, 0.3)$; $(0.5, 1.2, 0)$
B_1	$(0.4, 2.5, 0.3)$; $(0.35, 1, -0.2)$; $(0.25, 4, 0.05)$
B_2	$(0.6, 1.2, 0.4)$; $(0.4, 3.5, -0.15)$

All are smooth and monotone. B_1, B_2 are held-out profiles with recorded coefficient predictions before their scans. A_- tends to $v = +0.2$ on the right and is outside the nonpositive-flow assumption of the effective-horizon comparison. Three piecewise controls with a jump in v''' are excluded from the smooth-profile conclusions.

Section 7 uses $T_{0.5}$, $S_{2,1/4}$, A_+, M_4, B_2 and two further families that are used only for the low-frequency offset:

$$B_3:\ v = -1 + \sum_{i=1}^{2} A_i \Big[\operatorname{erf} \frac{x - s_i}{w_i} + \operatorname{erf} \frac{s_i}{w_i} \Big], \qquad B_4:\ v = -1 + 0.6 \tanh\Big(\frac{x}{0.6} - 0.7x^2 + 0.5x^3 \Big). \tag{11}$$

For B_3, $(s_i, w_i) = (0.35, 0.6), (-0.5, 1.0)$, and the amplitudes A_i are chosen so that the two terms contribute the fractions 0.55 and 0.45 of $v'(0) = 1$. Both are smooth and monotone, with $v \to -1.947$ and -0.101 (B_3) and $v \to -1.6$ and -0.4 (B_4) as $x \to \mp\infty$.

Independent numerical check. A second scattering code was written from Eqs. (2), (6) and (8) and the channel definitions alone, without the code, data or predictions behind the other results. It uses a different first-order formulation, a six-stage Gauss–Legendre integrator on short elements joined in one sparse linear system, and $\omega = 0.02, 0.01, 0.005, 0.0025$ at $\Lambda = 10, 20, 40, 80, 160$. Its value of b_0 is a quadratic fit of T_{ratio} in ω^2, extrapolated to $\omega = 0$, followed by a quadratic fit of $\Lambda^2[T(0)/T_{\mathrm{H}} - 1]$ in Λ^{-2} at $\Lambda = 20$–160. The profiles are $T_{0.5}$, B_4 and two further flows used only here,

$$N_1:\ v = -1 + \tfrac{1}{2} \tanh(2x + 0.4x^2 + 0.3x^3), \qquad N_2:\ v = -1 + A \Big[\operatorname{erf} \frac{x + 0.3}{0.7} - \operatorname{erf} \frac{0.3}{0.7} \Big], \tag{12}$$

with $A = 0.35\sqrt{\pi}\, e^{(3/7)^2}$, so that $\kappa = 1$. The values of Eq. (19), 0, 1.1336018, -0.6373333 and -0.4164931, were recorded before that computation and were not available to it. The computed values differ from them by -4.1×10^{-8}, -5.8×10^{-9}, $+3.1 \times 10^{-8}$ and -1.1×10^{-7}. Over 42 combinations of frequency and cutoff windows and polynomial degrees, the range of the computed b_0 contains the predicted value for B_4, N_1 and N_2; for $T_{0.5}$ the range $[-1.24 \times 10^{-6}, -5.7 \times 10^{-9}]$ misses 0 by 5.7×10^{-9}, less than the largest change of its central value between solver variants, 7.9×10^{-9}. These ranges measure the sensitivity of the extrapolation, not confidence intervals. The check is numerical agreement of Eq. (19) with an independent computation on four profiles; it does not prove A1–A3.

4 The local curvature coefficient from a known phase integral

Leonhardt and Robertson's Eq. (6) [5] gives the phase-integral temperature in the form

$$\frac{\omega}{T_{\rm phase}} = \frac{1}{i}\oint X_\omega(p)\mathrm{d}p, \tag{13}$$

with contour orientation chosen to give positive temperature. Put $u = v + 1$ and expand the inverse flow near the horizon as $x(u) = u/\kappa + x_2u^2 + x_3u^3 + x_4u^4 + \cdots$. The counter-propagating dispersion branch gives

$$u = \frac{\omega}{p} - \left(\sqrt{1 + p^2/\Lambda^2} - 1\right) = \frac{\omega}{p} - \frac{p^2}{2\Lambda^2} + \cdots . \tag{14}$$

At order Λ^{-2} only x_4u^4 contributes to the p^{-1} coefficient beyond the nondispersive term. Consequently,

$$[p^{-1}]X_\omega(p) = \frac{\omega}{\kappa} - \frac{2x_4\omega^3}{\Lambda^2} + \cdots , \qquad \frac{T_{\rm phase}}{T_{\rm H}} = 1 + 2\kappa x_4\frac{\omega^2}{\Lambda^2} + \cdots . \tag{15}$$

Writing $a = v''(0)$, $b = v'''(0)$, $d = v''''(0)$, inverse-series algebra gives

$$x_4 = -\frac{d}{24\kappa^5} + \frac{5ab}{12\kappa^6} - \frac{5a^3}{8\kappa^7}, \qquad \boxed{c_{\rm phase} = -\frac{d}{12\kappa^4} + \frac{5ab}{6\kappa^5} - \frac{5a^3}{4\kappa^6}.} \tag{16}$$

This derivative expression is a short formal consequence of the existing phase-integral theory, rather than a new temperature formula. In particular, $v + 1$ odd implies $a = d = 0$ and $c_{\rm phase} = 0$; $v''(0) = 0$ alone does not force $c_{\rm phase}$ to vanish.

Meaning of the expansion. Equation (15) is a coefficient extraction in a formal weak-dispersion Laurent expansion. At fixed nonzero ω, the expansion ω/p cannot stay close to the horizon inverse-flow series all the way to $p = 0$. The residue calculation is therefore an expansion in an appropriate overlap region, rather than permission to shrink an exact contour for an arbitrary nonlinear flow to that point. It describes the phase-integral approximation; applying the result to Eq. (2) requires control of amplitudes, connection errors and coupling to the co-propagating channel. The numerical measurements test this application on the specified backgrounds. They do not supply a general asymptotic theorem.

The effective-horizon coefficient to be compared. After identifying the conserved frequency denoted by Ω there with our ω, Eq. (133) of Del Porro, Liberati and Schneider [9] gives

$$c_{\rm EFH} = \frac{1}{4} - \frac{3}{8}\frac{v''(0)}{\kappa^2}. \tag{17}$$

For the two symmetric families this is 1/4, whereas Eq. (16) is zero. The asymmetric mixtures with zero second derivative provide a second useful separation: A_+ has $v''(0) = 0$ and $v''''(0) = -12$, so $c_{\rm phase} = 1$ at $\kappa = 1$. These comparisons isolate different proposed curvature coefficients while fitting the low-frequency intercept separately.

5 Measured curvature across profiles

Figure 1 compares all 19 smooth-profile extrapolations. The maximum absolute difference from Eq. (16) is 0.001269 after rounding upwards. The finite-cutoff data and extrapolations are both relevant: a three-point extrapolation can be less accurate than the largest-cutoff value, as F_4 illustrates.

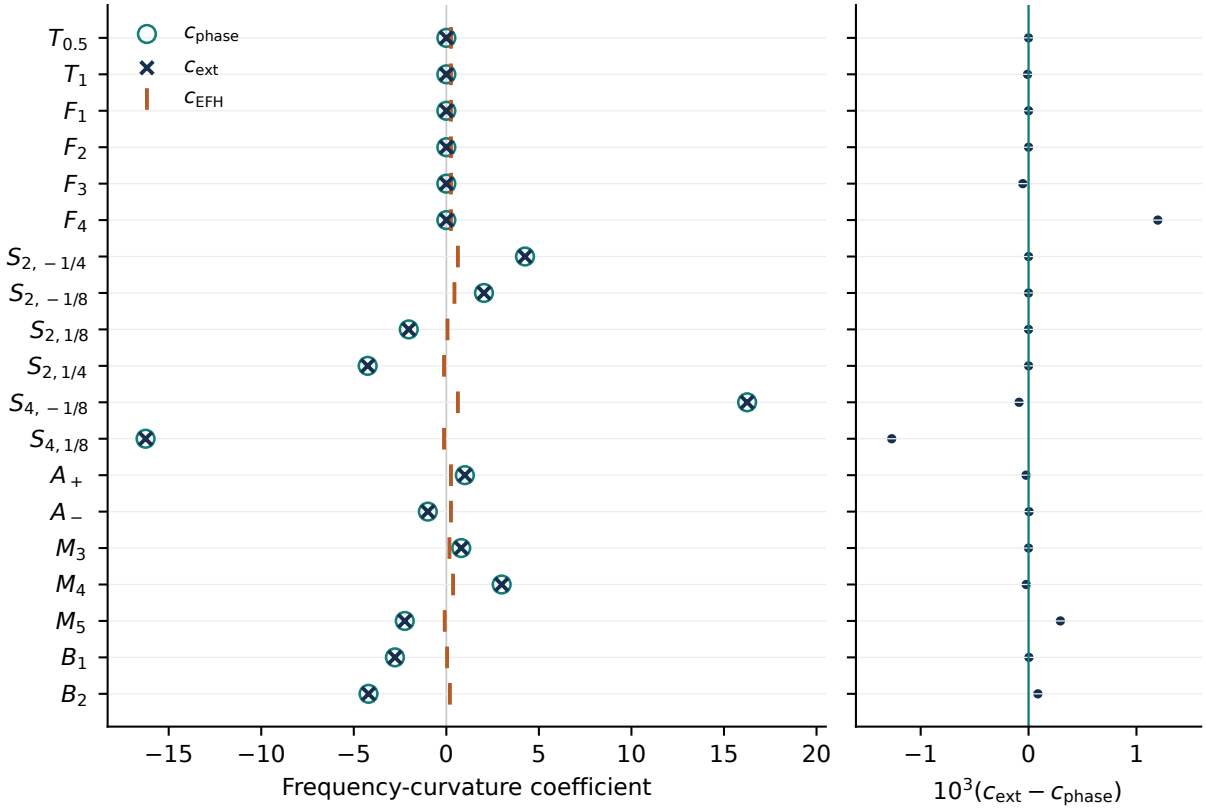


Figure 1: Smooth-profile curvature estimates. Left: the stored extrapolation c_{ext} (crosses), the phase-integral value c_{phase} (circles), and the effective-horizon coefficient c_{EFH} (ticks). Right: the difference $10^3(c_{\mathrm{ext}} - c_{\mathrm{phase}})$; no statistical error bars are assigned. All use $\kappa = 1$. The A_- result is shown as a scattering comparison, but lies outside the nonpositive-flow restriction of the effective-horizon treatment. Overlap of crosses and circles on the left is resolved by the residual panel.

Profile	$\Lambda_{\max}$	c_{ext}	c_{phase}	c_{EFH}
$T_{0.5}$	80	+0.000000	+0.000000	+0.250000
T_1	80	−0.000009	+0.000000	+0.250000
F_4	80	+0.001198	+0.000000	+0.250000
$S_{2,1/8}$	80	−2.031250	−2.031250	+0.062500
$S_{4,1/8}$	80	−16.251268	−16.250000	−0.125000
A_+	80	+0.999977	+1.000000	+0.250000
B_1	320	−2.778217	−2.778220	+0.040000
B_2	320	−4.203465	−4.203551	+0.191500

The table gives selected values; displayed decimal places transcribe the stored estimates and do not represent uncertainty bounds. The symmetric profiles approach a curvature near zero rather than 1/4. For the shifted and mixed profiles, both signs and the scale of the measured curvature follow Eq. (16). The largest residuals in the figure occur at $S_{4,1/8}$ and F_4, whose extrapolated discrepancies are approximately −0.001268 and +0.001198. Thus uniform agreement at the 10^{-4} level would overstate the whole-profile evidence.

6 Cutoff dependence and the low-frequency offset

The four profiles extended to $\Lambda = 320$ give the cutoff dependence in Fig. 2. The curves use the stored finite-cutoff coefficients, rather than an additional fit. Their approach to the phase-integral values supports the interpretation of c as the leading curvature coefficient for these examples.

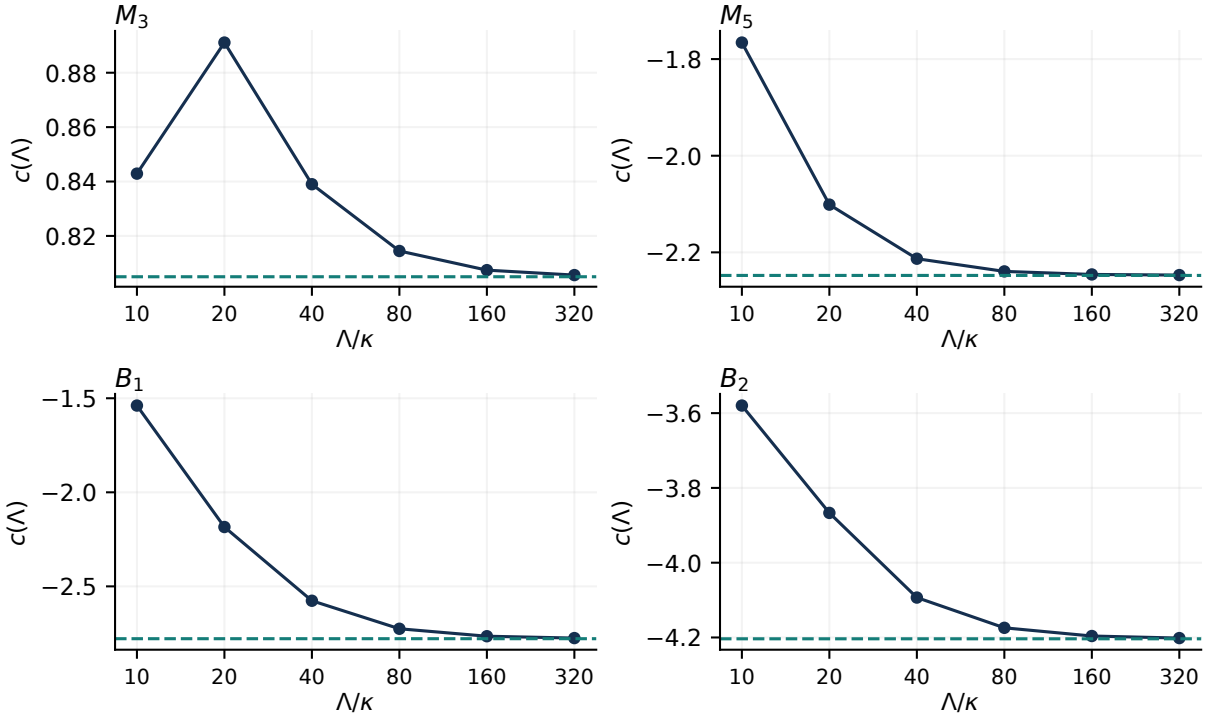


Figure 2: Cutoff dependence for two mixed and two held-out profiles. Points are the quartic-frequency-fit coefficients $c(\Lambda)$ from Eq. (9); dashed lines are Eq. (16). The logarithmic horizontal scale displays the six sampled cutoffs. The curves join stored measurements only. Mesh refinement was recorded at selected frequencies through $\Lambda = 160$; the last points have no such refinement check.

The low-frequency intercept from the frequency fits. In units $\kappa = 1$, the local formula derived formally in Section 7 is

$$\boxed{b_{0,\mathrm{loc}} = \frac{v''''(0) - 4v''(0)v'''(0) + 3[v''(0)]^3}{12}.} \tag{18}$$

For the held-out profiles the intercepts $b_{0,\mathrm{ext}}$ of the stored frequency fits, Eq. (9), are

Profile	$b_0(320)$	$b_{0,\mathrm{ext}}$	$b_{0,\mathrm{loc}}$
B_1	-0.819600	-0.815073	-0.815076
B_2	$+3.414518$	$+3.416860$	$+3.416864$

The absolute extrapolated differences are 2.93×10^{-6} and 3.69×10^{-6}, respectively. This is five-decimal-scale agreement, conditional on the extrapolation, rather than an established five-decimal error bound. The finite-320 offsets differ from Eq. (18) by about 0.00452 and 0.00235, so the extrapolation is material to the agreement. The leading phase integral in Section 4 produces no frequency-independent Λ^{-2} temperature term; Section 7 obtains this term from a separate zero-frequency calculation.

7 A formal derivation of the low-frequency offset

This section obtains Eq. (18) in two stages. Stage 1 reduces the $\omega \to 0$ limit of $T_{\rm ratio}$ to a zero-frequency problem without approximation in Λ. Stage 2 evaluates that limit by an expansion about the horizon turning point. In general units the result reads

$$\frac{T(0)}{T_{\rm H}} - 1 = \frac{b_0 \kappa^2}{\Lambda^2} + O(\Lambda^{-4}), \qquad b_0 = \frac{1}{12}\left[\frac{v''''}{\kappa^4} - \frac{4v''v'''}{\kappa^5} + \frac{3(v'')^3}{\kappa^6}\right] = \frac{g'''(0)}{12\kappa}, \tag{19}$$

with derivatives of v at the horizon, $T(0) = \lim_{\omega\to 0} T_{\rm ratio}$, and $O(\Lambda^{-4})$ the next order of the formal expansion. Here $u = 1 + v$ and $g(u) = v'(x(u))$, with $x(u)$ the local inverse near the horizon, so that $g(0) = \kappa$. If g is regarded as a function of v itself, the evaluation point is $v = -1$. Equivalently, the relative offset is $\kappa g'''(0)/(12\Lambda^2)$.

7.1 Hypotheses, order of limits and assumptions

The flow $v(x)$ is real, smooth, independent of Λ and monotone near the horizon. It tends to constants $v_- < -1$ as $x \to -\infty$ and $-1 < v_+ < 1$ as $x \to +\infty$, with derivatives decaying at least exponentially. The function $1 - v^2$ vanishes only at the simple zero $x = 0$; in particular there is no second turning point, such as $v = +1$. The scattering problem is nondegenerate, and the quantities $\mathcal{N}$ and $\mathcal{M}$ defined below satisfy $\mathcal{N} \neq 0$ and $\mathcal{N}\mathcal{M} > 0$. The limit $\omega \to 0$ is taken first at fixed Λ; the expansion in $1/\Lambda$ follows. No uniform interchange of the two limits is claimed. The calculation further uses:

(A1) *Low-frequency regularity and matching.* The outgoing-u solution, suitably normalized, has an expansion $f_0 + \omega f_1 + O(\omega^2)$ on bounded intervals, and the boundary-condition determinant has no exceptional zero at $\omega = 0$. There is an overlap region outside the flow transition but with $\omega|x| \ll 1$, in which this expansion is matched to the exterior soft waves and to the interior stiff waves.

(A2) *Oscillatory tails.* Integrals over the interior stiff oscillations are defined by Abel continuation (multiplication by $e^{\epsilon x}$ and $\epsilon \downarrow 0$), and this reproduces the finite-frequency boundary conditions. In particular, setting $\omega = 0$ before matching leaves no soft constant or slope absent at finite ω, so that the integration constants K_0 and K below vanish.

(A3) *Local evaluation.* To the order retained, the global integrals $\mathcal{N}$, $\mathcal{M}$ and the Wronskian W_c are given by termwise integration of the local turning-point expansion; contributions from regions away from the turning point are negligible to all orders kept. This includes the travelling-wave normalization: the formal second solution obtained by replacing Ai with Bi must combine with the first into the single incoming stiff component that defines W_c, to the order retained.

A1 and A2 are supported by the stage-wise numerical agreement below, and for constant interior velocity A2 reproduces exactly the primitives obtained from the finite-frequency stiff waves. None of A1–A3 is proved here. A proof would require a controlled global matching, for example a uniform Airy approximation near the turning point connected to WKB solutions away from it, with explicit remainder bounds.

7.2 Stage 1: reduction to a zero-frequency problem

Let $\phi_u^{\rm out}$ be the stationary solution containing, on the right, only the outgoing u wave, an incoming copropagating wave and the decaying evanescent solution, and on the left only the two incoming stiff waves. Two left-hand outgoing soft channels and the growing right-hand solution are excluded, which leaves one solution up to normalization. By pseudo-unitarity, the magnitudes of its incoming coefficients are those of the outgoing-u row of S. Write $\phi_u^{\rm out} = E\,(f_0 + \omega f_1 + \cdots)$ under A1 and $L_0 f = \Lambda^{-2} f'''' - [(1 - v^2)f']'$. Then Eq. (3) gives

$$L_0 f_0 = 0, \qquad L_0 f_1 = i\,[(v f_0)' + v f_0'']. \tag{20}$$

Integrating the first equation once, with vanishing constant K_0 (A2), gives $f_0' = \psi/\mathcal{N}$, where ψ is a real solution of

$$\psi'' = \Lambda^2 (1 - v^2)\,\psi \tag{21}$$

that decays as $x \to +\infty$. It is unique up to normalization, and all results below are invariant under rescaling of ψ. On the left, $\psi \simeq b\,e^{ikx} + b^*e^{-ikx}$ with $k = \Lambda\sqrt{v_-^2 - 1}$. With $\mathcal{N} = \int \psi\,\mathrm{d}x$ (Abel continuation on the left, A2) and $f_0 = \mathcal{N}^{-1}\int_{-\infty}^{x}\psi$, f_0 tends to 1 on the right and is a purely stiff oscillation on the left.

The once-integrated first-order equation is $\Lambda^{-2}f_1''' - (1-v^2)f_1' = i[vf_0 + \int_{-\infty}^{x} vf_0'] + K$ with $K = 0$ (A2). On the right, $f_1' \to -i(v_+ + I_v)/(1-v_+^2)$, where

$$I_v = \int v f_0'\,\mathrm{d}x = -1 - \delta, \qquad \delta = -\frac{1}{\mathcal{N}}\int (1+v)\,\psi\,\mathrm{d}x. \tag{22}$$

The exterior soft waves have $k_u = \omega/(1+v_+)+\cdots$ and $k_v = -\omega/(1-v_+)+\cdots$. Both have $|\Omega_c v_g| = \omega + o(\omega)$, so their leading flux normalizations agree, and the exterior solution is $\omega^{-1/2}[e^{ik_u x} + c_v e^{ik_v x}]$. Matching constant and slope terms in the overlap region of A1 gives

$$E = \omega^{-1/2}(1+c_v), \qquad c_v = \frac{1+I_v}{1-I_v} = -\frac{\delta}{2+\delta}. \tag{23}$$

On the left the stiff part of f_0 has amplitude $a = b/(ik\mathcal{N})$, and $\Omega_c v_g = k(v_-^2 - 1)$ at $\omega = 0$. Hence

$$|\beta|^2 \simeq |E|^2 F, \qquad F = |a|^2\Lambda(v_-^2-1)^{3/2} = \frac{|W_c|}{2\Lambda^2\mathcal{N}^2}, \tag{24}$$

where $W[f,h] = fh' - f'h$ and $W_c = W[\psi_c, \psi_c^*]$. Here ψ_c is the solution of Eq. (21) whose left asymptotic form is the positive-wave-number wave $b\,e^{ikx}$, so that $\psi = \psi_c + \psi_c^*$ and $W_c = -2ik|b|^2$, independent of x. Only $|W_c|$ enters below. Since $|\beta|^2 \sim 1/\omega$, $\log(|\alpha|^2/|\beta|^2) = \log[1 + (1-G)/|\beta|^2] = \omega(1-G_0)/C_\beta + o(\omega)$, with $C_\beta = \lim \omega|\beta|^2 = |1+c_v|^2F = 4F/(2+\delta)^2$ and $G_0 = \lim G = c_v^2 = \delta^2/(2+\delta)^2$. Therefore

$$\boxed{T(0) = \frac{C_\beta}{1-G_0} = \frac{F}{1+\delta} = \frac{|W_c|}{2\Lambda^2\mathcal{N}\mathcal{M}}, \qquad \mathcal{M} = \int(-v)\,\psi\,\mathrm{d}x = \mathcal{N}(1+\delta).} \tag{25}$$

Stage 1 makes no approximation in Λ; it holds under the hypotheses of Section 7.1, A1 and A2. The order-Λ^{-2} dependence on δ enters through $C_\beta = |1+c_v|^2F$; the factor $1/(1-G_0)$ changes $T(0)$ only at order Λ^{-4}, because $G_0 = O(\Lambda^{-4})$. With $T_n = \omega/\log(1 + 1/|\beta|^2)$ one has $T_n(0) = C_\beta = T(0)(1-G_0)$, so the two definitions share b_0. The first-order calculation determines $T(0)$ only. It does not show that $T_{\mathrm{ratio}}(\omega) - T(0)$ is of order ω^2; the even-power fits of Section 3 assume that form.

As a normalization check at leading order, replace the potential by its linearization $1 - v^2 \simeq 2x$ ($\kappa = 1$). Then $\psi = \mathrm{Ai}(sx)$ with $s = (2\Lambda^2)^{1/3}$, and $|W_c| = s/(2\pi)$, $\mathcal{N} = \mathcal{M} = 1/s$ give $T(0) = 1/(2\pi) = T_{\mathrm{H}}$. This checks the linearized problem only. For an exactly linear velocity, $1 - v^2 = 2x - x^2$, and $\mathrm{Ai}(sx)$ is not a solution.

7.3 Stage 2: expansion about the turning point

Write $u = 1 + v = x + a_2x^2 + a_3x^3 + \cdots$ with $a_n = v^{(n)}(0)/n!$ and $\kappa = 1$. Put $x = \varepsilon y$ with $2\Lambda^2\varepsilon^3 = 1$, so that $\Lambda^{-2} = 2\varepsilon^3$. Equation (21) becomes $\psi_{yy} = V(y)\psi$ with

$$V = \frac{2u - u^2}{2\varepsilon} = y + \varepsilon c_1 y^2 + \varepsilon^2 c_2 y^3 + \varepsilon^3 c_3 y^4 + \cdots, \quad c_1 = a_2 - \tfrac{1}{2},\ c_2 = a_3 - a_2,\ c_3 = a_4 - a_3 - \tfrac{1}{2}a_2^2. \tag{26}$$

With $\mathcal{N} = \varepsilon I_N$, $\mathcal{M} = \varepsilon I_M$ and $|W_c| = W_y[\psi_{\mathrm{Ai}}, \psi_{\mathrm{Bi}}]/(2\varepsilon)$, Eq. (25) becomes

$$\frac{T(0)}{T_{\mathrm{H}}} = \frac{\pi\,W_y[\psi_{\mathrm{Ai}}, \psi_{\mathrm{Bi}}]}{I_N\,I_M}, \qquad I_N = \int \psi\,\mathrm{d}y, \quad I_M = \int(1-u)\,\psi\,\mathrm{d}y. \tag{27}$$

Here $W_y[\mathrm{Ai}, \mathrm{Bi}] = 1/\pi$. For $y \to -\infty$, $\mathrm{Ai} \sim A\sin\theta$ and $\mathrm{Bi} \sim A\cos\theta$ with $\theta = \frac{2}{3}(-y)^{3/2} + \frac{\pi}{4}$ decreasing in y, so the positive-wave-number component is $\psi_c = \frac{1}{2}(\psi_{\mathrm{Ai}} + i\psi_{\mathrm{Bi}})$. This identification, which fixes the Wronskian factor, is the travelling-wave normalization contained in A3; the opposite choice of sign changes only the sign of W_c.

Each order is sought in the form $\psi_n = P_n\mathrm{Ai} + Q_n\mathrm{Ai}'$ with polynomials P_n, Q_n. The identity $(\partial_y^2 - y)(P\mathrm{Ai} + Q\mathrm{Ai}') = (P'' + 2yQ' + Q)\mathrm{Ai} + (2P' + Q'')\mathrm{Ai}'$ turns each order into a triangular recursion for

the polynomial coefficients. The only freedom is the normalization of ψ, which cancels in Eq. (27). Since Bi cannot be written in this form, the decaying condition on the subsonic side is built in. Replacing Ai by Bi with the same P, Q gives the formal second solution, and $\pi W_y = P^2 + PQ' - QP' - yQ^2$, which is independent of y at each order. The integrals use the Abel-regularized Airy moments $\int y^{3j}\mathrm{Ai}\,\mathrm{d}y = (3j)!/(3^j j!)$, with the other moments vanishing, and $\int y^n \mathrm{Ai}'\,\mathrm{d}y = -n\int y^{n-1}\mathrm{Ai}\,\mathrm{d}y$. The first correction is $\psi_1 = \frac{c_1}{5}(y^2\mathrm{Ai}' - y\mathrm{Ai})$.

With $P_0 = 1$ and no constant term added at higher orders, the ε^1 and ε^2 terms of πW_y, I_N and I_M vanish, and the ε^3 coefficients are

$$[I_N]_3 = -\tfrac{422}{225}c_1^3 + \tfrac{166}{45}c_1c_2 - \tfrac{16}{9}c_3, \qquad [\pi W_y]_3 = \tfrac{56}{225}c_1^3 - \tfrac{28}{45}c_1c_2 + \tfrac{4}{9}c_3, \qquad [\delta]_3 = -\tfrac{1}{2} - a_2, \tag{28}$$

with $[I_M]_3 = [I_N]_3 + [\delta]_3$. Combining them,

$$\Big[\frac{T(0)}{T_\mathrm{H}}\Big]_3 = 4c_1^3 - 8c_1c_2 + 4c_3 + \tfrac{1}{2} + a_2 = 4a_4 - 8a_2a_3 + 4a_2^3. \tag{29}$$

Since $\varepsilon^3 = \Lambda^{-2}/2$, this is $b_0 = 2a_4 - 4a_2a_3 + 2a_2^3$, i.e. Eq. (18); restoring κ gives Eq. (19). The expansion through ε^6, including b_0 and b_1 below, was reproduced by a second, independently written exact calculation that starts from the differential equation. The same calculation gives $\delta = -(1+v'')/(4\Lambda^2) + O(\Lambda^{-4})$, hence $G_0 \simeq (1+v'')^2/(64\Lambda^4)$ ($\kappa = 1$); in general units, $\delta = -(\kappa^2 + v'')/(4\Lambda^2) + \cdots$ and $G_0 \simeq (\kappa^2+v'')^2/(64\Lambda^4)$. The corrections to F and to δ share the term $-(1+v'')/4$, which cancels in $F/(1+\delta)$; neither quantity alone gives b_0.

Carried to ε^6 (derivatives through $v^{(7)}$), the expansion of $T(0)/T_\mathrm{H} - 1$ has nonzero terms only at ε^3 and ε^6: through this order it proceeds in powers of Λ^{-2}. The ε^6 term defines the next coefficient b_1 in $T(0)/T_\mathrm{H} - 1 = b_0/\Lambda^2 + b_1/\Lambda^4 + \cdots$ ($\kappa = 1$), a polynomial of 28 terms in the Taylor coefficients, or of 18 terms in the coefficients of g, Eq. (31) in Appendix A. Because $T_n(0) = T(0)(1 - G_0)$, b_1 refers to T_ratio, while b_0 is common to both definitions.

In terms of g, write $u' = g(u) = \kappa + g_1u + g_2u^2 + g_3u^3 + \cdots$. At $\kappa = 1$, $v'' = g_1$, $v''' = g_1^2 + 2g_2$ and $v'''' = g_1^3 + 8g_1g_2 + 6g_3$, so $b_0 = g_3/2 = g'''(0)/12$. For Riccati flows $u' = \kappa + \beta u + \gamma u^2$, which include T_D and all $S_{q,t}$, $g''' = 0$ and $b_0 = 0$. This does not make $T(0)$ equal to T_H at finite Λ, since $b_1 \neq 0$ in general.

7.4 Numerical checks of each stage

For seven profiles, Eq. (21) is integrated from the subsonic side inward, with the asymptotic tails added in closed form, to give $\mathcal{N}$, $\mathcal{M}$, W_c, F and δ. Independently, the boundary-value method gives $\omega \to 0$ limits of T_ratio, C_β and G as described in Section 3. Table 1 and Fig. 3 summarize the comparison.

Profile	b_0 (Eq. (19))	b_0 from scattering	difference	$\max\lvert\Delta T(0)\rvert/T_\mathrm{H}$	b_1 (local)	b_1 (fit)
$T_{0.5}$	+0.000000	+0.000000	-6.8×10^{-8}	1.1×10^{-11}	+0.25000	+0.24996
$S_{2,1/4}$	+0.000000	−0.000001	-1.1×10^{-6}	1.3×10^{-10}	+0.81250	+0.81245
A_+	−1.000000	−1.000000	$+1.6\times10^{-7}$	3.4×10^{-11}	+0.18750	+0.18723
M_4	−1.620000	−1.620090	-9.0×10^{-5}	1.6×10^{-11}	−53.41988	−53.43998
B_2	+3.416864	+3.417799	$+9.3\times10^{-4}$	2.0×10^{-11}	−239.75501	−240.02570
B_3	−0.607006	−0.607006	$+1.0\times10^{-7}$	1.2×10^{-11}	−2.24793	−2.24798
B_4	+1.133602	+1.133602	-1.4×10^{-8}	5.8×10^{-12}	+0.57353	+0.57350

Table 1: Low-frequency offset, $\kappa = 1$. Second column: Eq. (19) from the Taylor coefficients. Third: extrapolation of the (extrapolated) $\omega \to 0$ scattering limits $\Lambda^2[T(0)/T_\mathrm{H} - 1]$ at $\Lambda = 20, 40, 80$ in Λ^{-2} and Λ^{-4}; fourth: third minus second. Fifth: largest difference between the scattering limit and Eq. (25) over $\Lambda = 10$–80. Last two columns: b_1 from the local expansion and from a fit of the zero-frequency formula at $\Lambda = 28$–80 (relative tolerance 10^{-13}). B_3 and B_4, below the rule, are the prediction-first families.

Stage 1. Over all 28 cases ($\Lambda = 10, 20, 40, 80$) the extrapolated scattering limit, a four-point extrapolation from $\omega \geq 0.0025$, and Eq. (25) differ by at most 1.3×10^{-10} in $T(0)/T_\mathrm{H}$. Refitting the same stored solves on the three smallest or the three largest frequencies, quadratically in ω^2, changes the extrapolated value by at most 1.4×10^{-10}, while the value at the smallest frequency alone differs from Eq. (25) by up

to 2.7×10^{-7}. The pseudo-unitarity residual of the solves, below 1.9×10^{-9}, is a conservation diagnostic, not a bound on the temperature or extrapolation error. The limits G_0 agree with $\delta^2/(2+\delta)^2$ within a relative 8.4×10^{-7}. The agreement of C_β adds no independent content, since $T(0) = C_\beta/(1 - G_0)$ identically.

Stage 2. The difference between $\Lambda^2\delta$ and the local value $-(1 + v'')/4$ decreases, from $\Lambda = 40$ on, by a factor of three to four per doubling of Λ, to at most 4.8×10^{-3} at $\Lambda = 80$ and 3.0×10^{-4} at $\Lambda = 320$ (five profiles). The extrapolated b_0 agrees with Eq. (19) within 1.2×10^{-6} except for M_4 and B_2. For those two profiles, large Λ^{-6} terms limit the three-point extrapolation from $\Lambda \le 80$; the frequency-fit intercept of Section 6, which uses cutoffs up to 320, is 3.416860 for B_2. Subtracting b_1/Λ^2 from the zero-frequency value of $\Lambda^2[T(0)/T_\mathrm{H} - 1]$ at $\Lambda = 320$ reproduces b_0 within 6.3×10^{-6} (M_4) and 8.6×10^{-6} (B_2). The fitted b_1 agrees with the local expansion within 0.15% for all seven profiles.

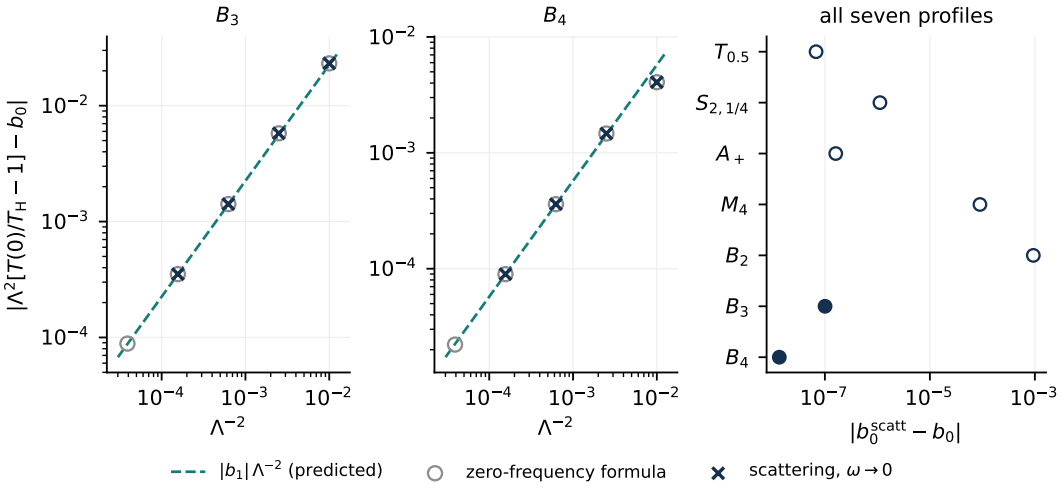


Figure 3: Low-frequency offset, $\kappa = 1$. Left and centre: for the prediction-first families, the deviation of $\Lambda^2[T(0)/T_\mathrm{H} - 1]$ from the predicted b_0, from the zero-frequency formula Eq. (25) (circles, $\Lambda = 10$–160) and from the scattering limits (crosses, $\Lambda = 10$–80). The dashed line is the predicted next term $|b_1|\Lambda^{-2}$, with no fitted parameter. Right: absolute difference between the extrapolated and the predicted b_0 for all seven profiles of Table 1; filled markers are the prediction-first families.

7.5 Prediction-first test

For B_3 and B_4, Eq. (11), the predictions of b_0, b_1 and δ were recorded before any zero-frequency or scattering calculation for these families. They used only the local formulas and Taylor coefficients from complex Cauchy integrals (two radii agree within 3.1×10^{-10} for B_3 and 8.4×10^{-8} for B_4). The success criteria were fixed at the same time: (1) b_0 extrapolated from the extrapolated scattering limits at $\Lambda = 20, 40, 80$, with basis $b_0 + e_1\Lambda^{-2} + e_2\Lambda^{-4}$, agrees with the prediction within $10^{-3}\max(1, |b_0|)$; (2) Eq. (25) agrees with the extrapolated scattering limit within 10^{-9} in $T(0)/T_\mathrm{H}$ at every computed Λ; (3) as a secondary test, b_1 from the zero-frequency formula at $\Lambda = 20$–160 agrees within 10%. All three are met for both families. The differences in (1) are $+1.0 \times 10^{-7}$ (B_3, predicted $b_0 = -0.6070063$) and -1.4×10^{-8} (B_4, $b_0 = +1.1336018$). For (2) the largest differences are 1.2×10^{-11} and 5.8×10^{-12}. For (3) the fits give -2.2601 and 0.5685 against the predicted -2.2479 and 0.5735; the refined fits of Table 1 give -2.24798 and 0.57350. With the alternative three-frequency windows of Section 7.4, the differences in (1) become $+1.0 \times 10^{-7}$ and $+1.1 \times 10^{-7}$ (B_3), and -1.3×10^{-8} and -1.3×10^{-9} (B_4).

Fixed before any zero-frequency or scattering result for these profiles were the profiles, the predictions, the three criteria, the cutoffs and basis of criterion (1), the frequencies and $\omega \to 0$ extrapolation bases of Section 3, the boundary-value discretization and the integrator tolerance. The cutoff list actually run, $\Lambda = 10, 20, 40, 80, 160$, is consistent with the criteria but was set at run time. Two choices were made later, with the results available. For (3), the basis is a least-squares fit of $\Lambda^4[T(0)/T_\mathrm{H} - 1] - \Lambda^2 b_0$ by $b_1 + e\Lambda^{-2} + e'\Lambda^{-4}$ at $\Lambda = 20, 40, 80, 160$, with b_0 fixed at its predicted value. The refined fits of Table 1 use the same basis at $\Lambda = 28, 40, 56, 80$, from a later run at $\Lambda = 10, 14, 20, 28, 40, 56, 80$ with tolerance 10^{-13}.

A first parameter choice for B_3 was rejected before any zero-frequency or scattering calculation, because its right-hand asymptotic velocity was positive. These are prediction-first tests by the same investigators, not blinded experiments. They differ from the held-out profiles B_1, B_2 of Section 3, which were held out for the curvature comparison; the b_0 comparisons for B_1, B_2 in Section 6 and for B_2 in Table 1 are not part of this protocol. This test does not replace a proof of A1–A3. The independent numerical check of Section 3 adds two further new profiles, N_1 and N_2.

7.6 Relation to previous work

The procedure of Stage 1 is that of Coutant and Weinfurtner [10, 11]. In their Korteweg–de Vries and Bogoliubov–de Gennes analyses the zero-frequency problem is reduced to a second-order equation, the solution that does not grow on the subsonic side is selected, and amplitudes are read from spatial integrals and asymptotic coefficients. In the Bogoliubov–de Gennes case the matching is carried to first order in the frequency, which separates the exterior counter- and co-propagating channels, and oscillatory integrals are given a regularized meaning. We transfer this procedure to Eq. (3); the integral weights and the form of Eq. (25) follow from the present operator, not from substitution into their coefficients. The Airy expansion about a turning point and the use of the inverse flow, as in Leonhardt and Robertson's Appendix B [5], are standard tools. Himemoto and Tanaka's Eqs. (5.2)–(5.5) [6] already allow Λ^{-2} corrections involving v'''', $v''v'''$ and $(v'')^3$, with coefficient functions left unevaluated and under different dispersion and connection conditions. What this section adds is the explicit Λ^{-2} coefficient, Eq. (19), for smooth single-horizon flows in the present model that satisfy the hypotheses of Section 7.1, as a formal result conditional on A1–A3. We have not found this coefficient in the sources examined, but cannot exclude an earlier statement elsewhere.

8 Precision, interpretation, and remaining analytical questions

What the two-method comparison establishes. The symmetric-tanh comparison contains 93 numerical table entries representing 64 distinct parameter points; repeated points occur in tables printed at different precisions. Of these entries, 79 pass every printed-field comparison using half the last printed unit, multiplied by 1.0001, plus 10^{-12}. The other 14 fail this criterion for a temperature shift; their largest excess over the printed half-unit is below 9.4×10^{-11} in $T/T_{\mathrm{H}} - 1$. All three above-cutoff entries agree. Absolute differences in coarsely printed T/T_{H} are below 5.0×10^{-7}; the digits of those tables limit the comparison. This is numerical compatibility at the recorded precision, with the small discrepancies stated, rather than exact agreement of every printed digit.

For this comparison set, the boundary-value method's maximum $\|S^\dagger \eta S - \eta\|_{\max}$ is below 8.8×10^{-12}, and its recorded refinement changes T/T_{H} by less than 6.0×10^{-13}. For the full smooth-profile scans the corresponding maximum norm residual is below 1.24×10^{-9}, the maximum recorded current variation below 1.84×10^{-9}, and 64 selected refined-mesh checks change the row temperature by less than 1.22×10^{-10}. The current variation is normalized by the larger of one and the maximum absolute current in a solution. These diagnostics test conservation and discretization at sampled points; they are not bounds on coefficient extrapolation. Multiplication by Λ^2 and extraction of a weak frequency curvature amplify small temperature errors.

Temperature definitions. The calculation also records the column ratio $T_{\mathrm{col}} = \omega/\log(|S_{u,+}|^2/|S_{P,+}|^2)$ and the occupation-number temperature $T_n = \omega/\log(1 + 1/|\beta|^2)$. In T_{col}, $+$ labels the positive-norm incoming high-wave-number channel and P the negative-norm outgoing long-wavelength channel on the left. With $N = |\beta|^2$, our principal definition is instead

$$T_{\mathrm{ratio}} = \frac{\omega}{\log[1 + (1 - G)/N]}. \tag{30}$$

Channel coupling must therefore be controlled at the order being compared; the three temperatures are not interchangeable by definition. Separating an offset and a curvature, as in Macher–Parentani Eqs. (42)–(45) [3], also prevents an intercept from being misidentified as a frequency correction.

How to interpret the effective-horizon comparison. The finite-scattering coefficients differ from a direct use of Eq. (17) on these backgrounds. Three analytical questions remain. First, the terms omitted in the approximant and adiabatic constructions [8, 9] need an error estimate at the order Λ^{-2}, rather than only at the leading thermal order. Second, one must compare an effective-horizon peeling slope with the logarithmic coefficient relevant to the normalized full-scattering ratio. Third, fixed-ω/κ weak dispersion and the subsequent $\omega \to 0$ limit need compatible, preferably uniform, control. These are possible explanations to investigate; none is identified as the cause by the present data. No claim that the effective-horizon analysis is erroneous follows from this comparison.

Conclusion. Full-scattering measurements over the specified smooth profiles support the curvature obtained from a known phase-integral formula. Two distinct numerical methods check the symmetric reference cases, while the wider profile set tests the derivative dependence with the boundary-value method. The low-frequency offset is obtained by a formal zero-frequency calculation, supported by stage-wise numerical checks, by two prediction-first families and by an independent computation on four profiles. The main analytical task that remains is to prove assumptions A1–A3 with a controlled remainder, for the stated class of flows and order of limits. Carrying the same matching to second order in the frequency might also give c within the full-scattering framework; this has not been attempted. More generally, broadened-horizon analyses [12, 13] motivate testing which aspects of a correction are local and which depend on the surrounding flow.

AI use. AI systems assisted with numerical implementation and analysis, the low-frequency asymptotic calculation and its exact algebra, an independent check of that calculation, literature comparison, algebraic exposition, English drafting, figures and consistency checks. The human author retains responsibility for the content and its stated scope; AI systems are not authors. The article has not undergone human expert review.

A The next coefficient b_1

With $\kappa = 1$ and $g(u) = 1 + \sum_{n\geq 1} g_n u^n$, the ε^6 term of the formal expansion gives the coefficient of Λ^{-4} in $T(0)/T_{\rm H} - 1$ as

$$\begin{aligned} b_1 = {} & \tfrac{1}{2}g_1g_2g_3 + \tfrac{7}{2}g_1g_5 + \tfrac{3}{4}g_1^2g_4 + 2g_2g_4 + g_3^2 + 3g_6 \\ & + g_1g_4 + \tfrac{1}{8}g_1^2g_3 + \tfrac{1}{2}g_2g_3 + g_5 + \tfrac{1}{8}g_1g_3 - \tfrac{1}{16}g_1^2g_2 \\ & + \tfrac{1}{4}g_4 - \tfrac{1}{8}g_1g_2 - \tfrac{1}{32}g_1^3 - \tfrac{1}{16}g_1^2 - \tfrac{1}{16}g_2 - \tfrac{1}{32}g_1. \end{aligned} \tag{31}$$

For $T_{0.5}$, $g = 1 - 4u^2$ and $b_1 = 1/4$. With g_n obtained from the Taylor coefficients of B_3 and B_4, Eq. (31) reproduces the predicted values of Table 1. Like b_0, Eq. (31) is a formal coefficient, conditional on A1–A3.

References

[1] S. Corley and T. Jacobson, "Hawking spectrum and high frequency dispersion," *Phys. Rev. D* **54**, 1568 (1996), arXiv:hep-th/9601073v1.

[2] W. G. Unruh and R. Schützhold, "On the universality of the Hawking effect," *Phys. Rev. D* **71**, 024028 (2005), arXiv:gr-qc/0408009v2.

[3] J. Macher and R. Parentani, "Black/White hole radiation from dispersive theories," *Phys. Rev. D* **79**, 124008 (2009), arXiv:0903.2224v3.

[4] A. Coutant, R. Parentani and S. Finazzi, "Black hole radiation with short distance dispersion, an analytical S-matrix approach," *Phys. Rev. D* **85**, 024021 (2012), arXiv:1108.1821v3.

[5] U. Leonhardt and S. Robertson, "Analytical theory of Hawking radiation in dispersive media," *New J. Phys.* **14**, 053003 (2012), doi:10.1088/1367-2630/14/5/053003.

[6] Y. Himemoto and T. Tanaka, "Generalization of the model of Hawking radiation with modified high frequency dispersion relation," *Phys. Rev. D* **61**, 064004 (2000), arXiv:gr-qc/9904076v3.

[7] A. Moreno-Ruiz and D. Bermudez, "Hawking temperature in dispersive media: Analytical and numerical study," arXiv:1908.02368v1 (2019), doi:10.1016/j.aop.2020.168268.

[8] F. Del Porro, S. Liberati and M. Schneider, "Tunneling method for Hawking quanta in analogue gravity," *C. R. Phys.* **25**, S2, 1–27 (2024; published online 24 Feb 2025), doi:10.5802/crphys.239, arXiv:2406.14603v2.

[9] F. Del Porro, S. Liberati and M. Schneider, "Hawking radiation with dispersion: reconciling the Bogoliubov and tunneling approaches," arXiv:2604.24861v1 (2026).

[10] A. Coutant and S. Weinfurtner, "Low frequency analogue Hawking radiation: The Korteweg–de Vries model," *Phys. Rev. D* **97**, 025005 (2018), arXiv:1707.09651v2.

[11] A. Coutant and S. Weinfurtner, "Low frequency analogue Hawking radiation: The Bogoliubov–de Gennes model," *Phys. Rev. D* **97**, 025006 (2018), arXiv:1707.09664v2.

[12] S. Finazzi and R. Parentani, "Spectral properties of acoustic black hole radiation: Broadening the horizon," *Phys. Rev. D* **83**, 084010 (2011), arXiv:1012.1556v2.

[13] A. Coutant and R. Parentani, "Hawking radiation with dispersion: The broadened horizon paradigm," *Phys. Rev. D* **90**, 121501 (2014), arXiv:1402.2514v2.